

Roopak Bhukya

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Professional Summary

AI/ML Engineer who ships production AI systems end-to-end, from model development to deployment. Built **AI-powered computer vision proctoring** and **AI-based KYC verification** systems used in production, **LLM-powered agentic pipelines** for market intelligence, and a **real-time AI voice agent** for primary care. Experienced in LLM fine-tuning, RAG, multi-agent systems, and MLOps, owning the backend, AI workflows, and infrastructure from prototype to production.

Education

B.Tech, Materials Science & Metallurgical Engineering

2021 – 2025

Indian Institute of Technology (IIT) Hyderabad

Professional Experience

AI Full Stack Developer

Nov 2025 – Jul 2026

Net Connect Global Pvt. Ltd., Bengaluru

- Built and productionized a **multimodal AI proctoring platform** using computer vision and AI agents to detect faces, voices, gaze, phones, keystroke patterns, and browser activity; reached **93% detection accuracy** on a **500+ image benchmark** of simulated cheating scenarios, generating **30+ risk reports daily**.
- Cut per-interview processing cost by **50% (INR 8 to INR 4)** by migrating proctoring workloads to **AWS EC2 Spot Instances** and parallelizing analysis with **ThreadPoolExecutor**; reduced report time for 30-minute sessions from **5.5 to 4 min**.
- Developed a production **KYC pipeline** for **1,000+ Aadhaar/PAN verifications**, combining OCR, OpenAI-based resolution of unclear fields, and document-to-face matching to reduce manual checks; deployed on AWS EC2 with GitHub Actions CI/CD.

Machine Learning Intern

Jun 2025 – Sep 2025

Mobius by Gaian, Hyderabad

- Improved **knowledge-graph link-prediction** performance by **33%** by migrating from R-GCN to CompGCN; productionized a reusable **Kubeflow Pipelines** workflow covering GNN training, evaluation, experiment tracking, and model versioning.
- Fine-tuned **Mistral LLMs** on proprietary domain datasets using PEFT techniques, delivering a **1.4 percentage-point accuracy gain** while reducing inference latency by **300 ms**.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript

AI / ML: PyTorch, Scikit-Learn, TensorFlow, PEFT, LoRA, NLP, Computer Vision, LLM Fine-Tuning, OCR

LLM / GenAI: RAG, Embeddings, Hybrid Search, Agentic AI, LangChain, LangGraph, Qdrant, vLLM, Groq, OpenAI API

MLOps & Cloud: Kubeflow, MLflow, Docker, GitHub Actions, AWS, Logfire, OpenTelemetry

Backend / Web: React, Next.js, FastAPI, Flask, REST APIs, WebSockets, Twilio, Deepgram

Databases & Tools: PostgreSQL, MySQL, Git/GitHub, Jupyter, Postman, Playwright, pytest, Firecrawl, Tavily

Projects

Clinexa: RAG-Powered Healthcare Voice Agent | GitHub

Sep 2026 – Oct 2026

- Engineered a safety-aware RAG pipeline over 4,200+ clinical-guideline chunks using Qdrant, BGE embeddings, BM25/RRF hybrid search, cross-encoder reranking, and age- and pregnancy-aware filters at **121 ms p95 retrieval latency**; built a phonetic sparse-vector search over **252K+ medicine names** with **87% recall@10** on 4,000 synthetic noisy queries.
- Built a real-time voice assistant in Python using FastAPI WebSockets, Twilio Media Streams, and Deepgram streaming STT/TTS, with retrieval tools at **~300 ms p95 latency** and OpenTelemetry tracing in Pydantic Logfire; developed an evaluation harness for Hits@k, MRR, and NDCG, backed by **356 automated tests**.

Multi-Agent Market Intelligence System | GitHub

Aug 2026 – Sep 2026

- Engineered a **multi-agent RAG system** with LangGraph, LangChain and Groq, orchestrating **9 agents and 25 tools** in parallel and using Qdrant hybrid search (dense + BM25, RRF, cross-encoder reranking) over **25,497 chunks** spanning 20 years of NIFTY 50 data to automate daily market intelligence reports.
- Built a Firecrawl and Tavily news ingestion and web scraping pipeline across 8 sources, cutting thin or missing article summaries from **25% to 0%** with full-text extraction; hardened it with a **3-tier LLM fallback chain**, Logfire tracing, a fact-checking agent, and **137 automated pytest tests**.

Certificate Verification Platform | GitHub

Jul 2026 – Aug 2026

- Built an **asynchronous certificate-verification API** with Python and FastAPI that runs PaddleOCR, EasyOCR, and Tesseract in parallel with consensus voting to improve extraction accuracy, then uses Playwright to cross-check certificate details against official records from **72 certification providers**.
- Offloaded difficult field extraction to Mistral and Qwen **vision-language models** running in the background, reducing processing time by up to **2 s per document**; added TruFor image forensics to detect tampered or forged certificates.